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RESEARCH

DAOs as property owners: a conceptual exploration from the perspective of organizational system theory

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Abstract
The development of Bitcoin and its underlying technology blockchain has enabled a new phenomenon called *decentralized autonomous organizations* (DAOs). DAOs can be perceived as self-governing organizations whose management is based on programmed and encoded rules on a decentralized and distributed peer-to-peer network. These DAOs typically manage and allocate funds, often in the form of cryptocurrencies. However, in recent years, a variety of DAOs have been established to provide services (e.g., currency exchange, project financing), curate collections (e.g., art collections), or own and manage real assets (e.g., land). Currently, DAO literature focuses mainly on online communities managing digital assets; however, DAOs owning physical properties differ from them in localized communities, asset indivisibility, and additional complexity in collective acquisition, ownership, limited physical capacity, and decentralized governance. Such property-owning DAOs are interesting, because they fuel the transition from purely online organizations into organizations integrating with the physical world. From an organizational system theory perspective this article explores how a DAO owning properties could be designed by exploring three DAO projects that own properties. Applying a *conceptual research design*, we first identify *DAO Design Principles* obtained by traditional organizational system theory, followed by examining and describing the core organizational principles for property-owning DAOs. Based on a comprehensive discussion of the conceptual findings, we present a research agenda for further studies on DAOs owning properties.

Keywords Decentralized autonomous organization · Blockchain · Organization theory · Governance · Community · Research agenda

Introduction

In a decentralized autonomous organization (DAO), pre-programmed rules are executed through smart contracts (Murray et al. 2021). All administrative and functional rules of a DAO rely on the decision-making of the community, with these rules being stored on a tamper-proof blockchain (Wang et al. 2019). The idea of such blockchain-based, autonomous and decentralized organizations was discussed among blockchain developers as early as 2013 (Buterin 2013). At that time, some developers were still clearly of the opinion that DAOs had to be organized and function without any human involvement and that governance—i.e., the development, administration, and control of the DAO—could be carried out solely via programs stored on a blockchain (Larimer 2013). However, most developers agreed that human involvement is still crucial and the term DAO in this context originates finally from a blog by Ethereum founder Buterin (2014) describing a DAO "as an entity that lives on the Internet and exists autonomously, but also relies heavily on individuals to take on certain tasks that the autonomous entity cannot do itself."

In line with this conceptual understanding, Hassan and De Filippi (2021) also define a DAO as "a blockchain-based system that enables people to coordinate and govern themselves mediated by a set of self-executing rules deployed on a public blockchain, and whose governance is decentralized (i.e., independent from central control)." In an extensive literature review, Santana and Albareda (2022) demonstrate that a DAO as a blockchain-based organization is based on

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three specific principles: (i) on decentralized governance without formal leadership and decision-making hierarchies, (ii) on automated processes that are programmed and executed using smart contracts as coded rules on a decentralized blockchain, and (iii) on an autonomous and dynamic community formed by autonomous and independent members.

Since 2014, quite a few DAOs have emerged, which collectively manage a treasury (especially cryptocurrencies), particularly in decentralized finance.[1] A well-known example is MakerDAO, whose members can use their governance token to ensure the efficiency, transparency, and stability of the stablecoin DAI. Or Uniswap, a decentralized exchange protocol (DEX) controlled and managed by UNI token holders. The vast majority of these DAOs hold their treasury in their own tokens. These tokens may be used to vote in the governance processes, but at the same time, they are traded on the market. This leads to the fact that voting power can be bought and as a result, many DAOs suffer from concentrated voting dynamics (Feichtinger et al. 2024). Practice and literature have come up with elaborated voting mechanisms to provide the basis for a more equal voting power distribution. A good overview on the different mechanisms is provided in Spychiger et al. 2025. While sophisticated voting mechanisms may improve voting outcomes in DAOs, designing a DAO involves several other elements. Current literature at the intersection of organizational theory and decentralized governance focuses mainly on decentralized cryptocurrency platforms (Hsieh and Vergne 2023), enterprise blockchains (Goldsby and Hanisch 2022), and blockchain technology as both a governance mechanism (Lumineau et al. 2021) and an enabler for the emergence of decentralized platforms (Vergne 2020). It is generally understood that unlike traditional platforms, which can unilaterally change rules and disrupt the value generating processes, DAOs provide foreseeable and neutral value dynamics which are embedded into the platform rules onto the blockchain (Spychiger et al. 2025). This participatory value management not only makes DAOs more attractive for contributors, but also could provide an effective organization of open-source communities, providing a fairer and more resilient alternative to centralized business models (Vergne 2020; Hsieh and Vergne 2023). Recently, Lustenberger et al. (2024) proposed a DAO-specific framework for the organizational design encompassing three constituent elements (*purpose, legislation, and use-case*) and five structural elements (*organizational bodies, decision-making, treasury, tokens, communication*).

However, these frameworks address challenges in DAOs governing digital assets. There are, though, some projects that implement DAOs owning real-world properties, such as land or buildings. These DAOs—even though currently in the minority—constitute an evolutionary step of these blockchain-native organizations as they venture into the physical world. They often form around a purpose that specifically involves a property, e.g., the collective acquisition, ownership, and governance of a property. Like any DAO, property-owning DAOs enable decentralized, transparent and collective management of physical assets, reducing the need for intermediaries and improving participation in decision-making. They provide a more efficient and equitable approach to property ownership then traditional ownership forms, minimizing inequality and ensuring that no single entity has control over wealth distribution. A very well-known example is CityDAO,[2] a DAO that owns a piece of land in the USA. The community (the token holders) collectively own this piece of land and can also vote on its use. In CityDAO Proposal 14, the community decided that the land should be used as a nature preserve. Similar to CityDAO, the Cabin[3] project is trying to build a community that uses a DAO to manage properties, such as land, houses and even villages. However, Cabin also emphasizes real-life gatherings to create local communities besides the online participation through blockchain-based voting. In 2019, Dezentrum—a think tank in Switzerland—developed the concept of no1s1 (read as "no one's one") that should become the first house that owns itself and decides on its own use. With the help of smart contracts, no1s1 should be able to manage itself. Unlike most existing DAOs that operate exclusively in digital environments and manage virtual assets, CityDAO, Cabin, and no1s1 are designed to make a specific physical space the object of participation and community. Building on the characteristics of a DAO, these projects are, therefore, not only able to create and maintain a physical space itself, but also to own it.

While the design of DAOs still leaves many questions open (Tan et al. 2023), the element of *decentralized governance* in DAOs currently causes the most discussions and concerns in the literature. Atzori (2015), for example, early on argued that coordination by responsible, centralized leadership is a necessary condition for reliable governance of an open and democratic community. However, Reijers et al. (2021) clearly believe that DAOs cannot be equated with traditional democratic communities, as DAOs have a much more radical voluntary character. Thus, each member is free to leave the organization at any time and to create a new DAO community through a *hard fork* if needed. This possibility (or threat) of a technical copy of a DAO can provide a possibility to overcome conflicts by dividing the community (Ziolkowski 2021). However, a recent article by Heo and Yi (2023) states that the power to make important decisions

[1] https://deepdao.io/organizations.

[2] https://www.citydao.io/.

[3] https://www.cabin.city/.

in blockchain-based governance structures often ends up in the hands of a small group of founders, key developers or powerful validators, and the community often has less to say than commonly assumed.

A DAO governing a property adds an additional layer of complexity. It is not only about coordinating an online community deciding on the allocation of digital assets, rather the community becomes fragmented and split up in local subgroups being connected through their proximity to the local property. Nevertheless, the community still decides on certain aspects of the localized properties as a whole. This duality between the global and local community creates even more challenges for the organizational design of the DAO. Furthermore, forking the community and copying the assets is no longer possible due to the indivisibility of the assets. Therefore, this important conflict resolution measure cannot be applied by DAOs owning properties. Yet, academic literature neglected these types of DAOs so far and there are no clear frameworks for the organizational design of DAOs owning properties that could be applied by projects. Given that such DAOs have the potential to enable the transition for DAOs from a purely online organization to an organization integrating the real world, it seems important to develop guidance on how to design such organizations.

Many DAOs reinvent organizational structures; however, there is a long-standing strand of organizational design literature that may provide important insights for DAOs owning properties. One possibility would be to focus on newer organizational concepts related to information technology, such as 'virtual organization', 'meta-organization' or 'online community'. However, all these concepts are mainly concerned with some specific aspects of organizations. The 'virtual organization' attempts to minimize switching costs between alternative requirement satisfiers using information technology (Mowshowitz 1997), the 'meta-organization' describes organizations whose agents are legally autonomous along the two dimensions of stratification and membership boundaries (Gulati et al. 2012), and the 'online community', finally, sees a group of people discussing online as a way of developing relationships and transferring knowledge within and between organizations (Butler et al. 2014). While these concepts can provide some valuable insights into the organizational structures of DAOs, they are not able to capture the design of DAOs from a comprehensive organizational understanding, as established foundational organizational theories allow. One of these foundational theories is the organizational system theory. According to this perspective, the core of the organizations are decision-making processes and the associated autonomy in these processes to address complexity and uncertainty (Luhmann 2000). As such, the organizational system theory perspective is a good fit to explore organizational design for DAOs, where decision-making and autonomy are pivotal (Schädler et al. 2023).

Leveraging on organizational system theory, the following research question will be answered:

How can the organizational structures of a DAO owning properties be designed?

Following a conceptual research design (Meredith 1993; Jaakkola 2020), our contribution to academia and practice is twofold: *first*, we combine the traditional organization system theory with the relatively young literature on DAOs to synthesize *DAO Design Principles*. By this, we make use of already existing knowledge of organizations and DAOs and apply that understanding to a new domain. *Second,* we suggest a concept on how the organizational structures of DAOs owning properties can be designed by applying the DAO Design Principles to three property-owning DAO projects. Through this, we provide a conceptual foundation for any further exploration of the organizational design of these new organizations governing properties.

In the following, "Research design" section outlines the research design followed by this study. "Model development: DAO design principles" section presents the literature on organizational system theory and on DAOs to identify the main *DAO Design Principles*. In "Empirical analysis" section, we shortly introduce the three property-owning DAO projects and apply the previously identified *DAO Design Principles*. In "Discussion" section, the findings for DAOs owning properties will be discussed and an extensive research agenda will be provided in "Research agenda" section. Finally, "Conclusion" section concludes.

Research design

Conceptual research, as outlined by Gilson and Goldberg (2015), focuses on integrating and linking existing theories in novel ways to expand our understanding, rather than relying on empirical data or proposing entirely new theories. While Fulmer's (2012) review of award-winning *Academy of Management* articles found no single ideal structure for such research, Meredith (1993) and Jaakkola (2020) offer valuable methodological approaches. These approaches focus on synthesizing arguments by combining previously developed concepts and theories, which represents a useful methodological framework for this study. In accordance with Meredith (1993), we aim to develop conceptual models by describing the phenomena as well as identifying and integrating relevant theoretical elements to extend the current understanding and to create new knowledge through our research. In line with the conceptual research approach of Jaakkola (2020), we start from an interesting *focal phenomenon*—in our case DAOs owning properties—and analyze and evaluate this phenomenon with the help

of the organizational system theory (Kühl 2021; Luhmann 2000; Meibach 2007; Mintzberg 1983). Our focus lays on the organizational concepts derived from the system theory, as we believe that these concepts provide a promising theoretical approach to analyze dynamic social systems like DAO communities. In accordance with Jaakkola (2020), the organizational system theory is, therefore, our *method theory* which we apply to the *domain theory* of DAO. From a systems theory perspective, organizations can be understood primarily through their decision-making processes and the autonomy they hold in making those decisions to cope with complexity and uncertainty (Luhmann 2000). This focus on decision-making aligns with the growing recognition of decentralized decision-making as a key driver for participation in blockchain-based systems like DAOs (Schädler et al. 2023). By integrating concepts from the method theory with the DAO research domain, we can develop a comprehensive conceptual model that we can then apply and evaluate within different property-owning DAO projects, such as no1s1, CityDAO, and Cabin.

In developing our conceptual model with a problem-focused approach, we pay particular attention to ensure that the line of argument is coherent and transparent to the reader. Following Jaakkola (2020), as shown in Fig. 1, we first use concepts from organizational system theory to integrate them with the DAO literature, aiming to develop a model consisting of *DAO Design Principles*. In a second step, we present three property-owning DAO projects to apply, evaluate and discuss the established *DAO Design Principles* in more detail. Finally, with the guiding structure provided by the application to three property-owning DAO projects, we can discuss our conceptual findings, "highlight directions for future inquiry" (Gilson and Goldberg 2015) and lay down the conceptual base for empirical studies including real-life implementations.

Model development: DAO design principles

Organizational design principles from a systems theory perspective

Organization can be described as a group of people working together in an organized way towards a common goal (Vahs 2019). Kühl (2021) further argues that organizations as a particular form should be clearly distinguished from other social systems, such as family, groups, networks, protest movements or nation states. In this regard, a particular reference can be made to the systems theory perspective of Luhmann (2000), who states that organizations consist of decisions, which are able to reduce organizational complexities and uncertainties in the pursuit of goal attainment due to self-defined formalities and structures. In contrast to other social systems, organizations must thereby fulfill a clear purpose and thus also pursue clear goals with their activities (Luhmann 2000). In this respect, according to Kühl (2021), organizations are not democratically organized like modern society, but clearly hierarchical, because despite today's dehierarchization and decentralization rhetoric, hierarchies are still considered necessary in science and practice, especially in complex organizations (Miebach 2007). In his systems theory understanding, Luhmann (2000) further states that in organizations power is given to those actors that are able to reduce uncertainties for their subordinates through their decisions.

Furthermore, organizations are characterized by the fact that their members (e.g., employees) are selected and their role in the organization is formally defined in terms of duties and responsibilities (Luhmann 2000). Thus, it is always clear at a certain period of time, who belongs to an organization and who does not, and what decision-making power someone formally has. In this context, it is the organization itself that can decide on members joining—or leaving—and the respective responsibilities and duties of each of member. Members that do not comply with these organizational rules can be pressured to quit their membership (Kühl 2021). Hence, it can be assumed that members of an organization are implicitly impacted by the accountability of their own actions.

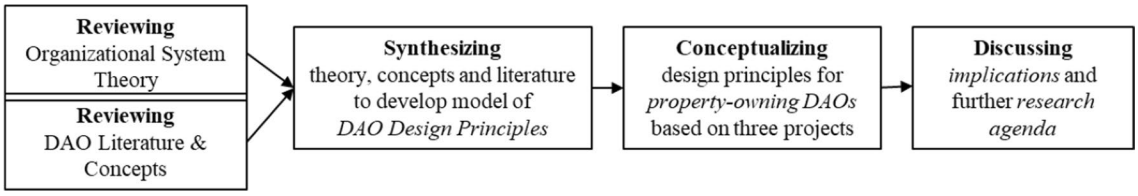

Fig. 1 Conceptual research design

However, as Meibach (2007) further points out—and Luhmann (2000) had already agreed in principle—strictly hierarchical decision-making and power structures are often not sufficient to ensure complex processes and uncertainty absorption in organizations. Therefore, alternative forms of decision-making and power can often be found in organizations that are not based on the traditional vertical, hierarchical power chains found in autocracies and bureaucracies. Mintzberg (1983), for example, defines ideology, expertise, and politics as decision-structuring forces in organizations in addition to hierarchical authority. Based on a common ideology, for example, decision-making power can be distributed without conflict in organizations, according to Mintzberg (1983), because the strong conviction of each individual of the common mission creates trust among organizational members. In turn, in organizations structured as meritocracies, decision-making power is defined by the recognized expertise and professional competence of individual members. Finally, according to Mintzberg (1983), politics, as a conflicting and decisive factor, takes over decision-making power in organizations when uncertainties arise that cannot be absorbed by either a formal authority (*bureaucracy, autocracy*), a commonly accepted ideology (*mission*), or the confirmed expertise of individual members (*meritocracy).*

In a later article, Mintzberg (1984) further elaborated that in organizations a particular power structure is dominant at each stage during their life cycle of formation, growth, stagnation, and decline. Based on Mintzberg's model of *Organization Life Cycles* (Mintzberg 1984), it can be said that organizations are often founded as autocracies and evolve into bureaucracies (hierarchy), missions (ideology), or meritocracies (expertise) throughout their life cycle. These different decision-making and power structures in organizations are grouped under the concept of *governance* in recent literature. For example, Bevier (2012) states that governance refers to all processes exercised by leadership over other members of a social system based on rules, norms, or power. Hufty (2011) further opines that governance must be understood as all those processes of interaction and decision-making among actors that lead to the creation, reinforcement, or reproduction of social norms and institutions. In this respect, the decision-making power structures listed by Luhmann (2000) and Mintzberg (1983) and thus, the different forms of uncertainty absorption in organizations, can be understood as *governance structures,* while in turn the process of exercising domination and decision-making power in organizations can be described as *governance*.

Based on the above collected organizational system theory literature we can summarize that organizations are grounded on a set of key principles that are constitutive to the organizational design. As outlined in Table 1, we define *purpose*, *membership,* and *governance* as the three key design principles of an organization. These three design principles can be further divided into six sub-design principles. Each formal organization, therefore, needs to decide about (i) its goals and objectives, (ii) its structures and rules to reach these goals, (iii) the entry and exit of its members, (iv) the formalized duties of its members and who to keep them accountable for their actions, (v) the delegation of decision-making to some of its members according to an accepted governance model, and finally, (vi) the way to implement flexibility to evolve over the life cycle.

DAO design principles from a system theory perspective

Decentralized autonomous organizations (DAOs) mark a profound shift from traditional organizational models (Vergne 2020). Analyzing DAOs through the six design principles established from a systems theory perspective provides insight into how these novel entities differ from conventional organizations.

Goal: shared goal by community

In the context of DAOs, the design of goals is fundamentally different from traditional organizations due to the decentralized and digital nature of DAOs. The goal of a DAO is typically established and pursued by a community of members who share a common interest (El Faqir et al 2020).[4] This shared purpose is crucial for aligning the activities of the DAO with its intended outcomes. The importance of a well-defined goal is particularly evident in DAOs, as the often anonymous and geographically dispersed membership can lead to situations, where members pursue contradictory objectives, potentially undermining the organization's coherence (Beck et al. 2018). Therefore, it is essential that the purpose of a DAO is not only clear but also widely accepted and internalized by the member community. This shared purpose ensures that the DAO operates in a cohesive and directed manner, with all activities aligned towards achieving the established goals (Mintzberg 1983).

Structure and rules: incentive structure

Like in traditional organizations, structures and rules within DAOs are designed to align member behavior with the organization's goals, predominantly through a formal incentive structure (Faqir-Rhazoui et al. 2021). However, unlike

[4] The Ethereum Community defines a DAO as "a collectively-owned organization working towards a shared mission" (https://ethereum.org/en/dao/).

Table 1 Evolution and synthesis of design principles for property-owning DAOs

Purpose	Membership	Governance
Organization		
Clear goal Clear purpose is pursued with organizational activities (Luhmann 2000)	*Defined entry & exit* Formal selection of members; boundaries are defined clearly (Kühl 2021)	*Delegated decision-making* Reduction of uncertainties by delegation of decision-making based on hierarchy, ideology or expertise (Luhmann 2000; Mintzberg 1983)
Structure & rules Purpose is reached through definition of structures and rules (Vahs 2019)	*Formalized roles & accountability* Roles are formalized by duties and responsibilities (Luhmann 2000); Members remain accountable (Kühl 2021)	*Traditional life cycle* Evolution of power structures along the life cycle from autocracies to bureaucracies (hierarchy), missions (ideology) or meritocracies (expertise) (Mintzberg 1984)
DAO		
Shared Goal Shared purpose is pursued through a community (El Faqir et al. 2020)	*Autonomous Entry & Exit* No selection of members; Boundaries are defined by tokens (Faqir-Rhazoui et al. 2021)	*Decentralized decision-making* Reduction of uncertainties by decision-making through on-chain processes (Wright 2021)
Incentive structure Purpose is reached through aligning the behaviors of members with tokens (Faqir-Rhazoui et al. 2021)	*Informal roles & lack of accountability* No formalized roles; anonymous members are not accountable (Rikken et al. 2019; Braun et al. 2021)	*Decentralization life cycle* Evolution of power structures along the life cycle from off-chain autocracies to on-chain disempowerment (decentralization) or division (forks) (Ziolkowski et al. 2020)
Property-owning DAO		
Shared Property Usage A shared and self-owned property is provided for the use of a community	*Entry & exit by usage* Members are users of the property; Boundaries are defined by a non-accumulative usage-token	*Community self-administration* Reduction of uncertainties through on-chain self-administration by the community of property users
Use-based incentive structure Participation, usage, and maintenance of the property is incentivized through different tokens	*Task-based roles & limited accountability* DAO members take on tasks for the DAO autonomously; Limited accountability through physical presence	*Participatory life cycle* Evolution of power structures along the life cycle from property owner to on-chain community of users. Survival of the DAO is based on usage

traditional organizations, where hierarchical controls and rules are imposed, DAOs utilize smart contracts on a decentralized blockchain to enforce rules autonomously (Hassan and De Filippi 2021; Lumineau et al. 2021). These smart contracts encode the organization's rules and structures in a transparent and immutable manner, reducing uncertainties and ensuring compliance without the need for centralized authority (Wright 2021). As Faqir-Rhazoui et al. (2021) further highlight, the incentive structure in DAOs typically involves the distribution of tokens, which serve as both a medium of exchange and a mechanism for aligning member behavior with organizational objectives. Positive and negative incentives, encoded within the DAO's smart contracts, encourage members to engage in activities that support the DAO's goals and discourage behavior that could harm the organization. This system of incentives is crucial for maintaining the integrity of the DAO's purpose and ensuring that all members are working towards the same objectives.

Member entry and exit: autonomous

In the traditional view, the boundaries of the organization must be clearly defined by established membership rules (Kühl 2021). It must always be clear who belongs to an organization and who does not, and the rules for (later) joining or leaving must also be clearly defined. However, in DAOs, the processes for member entry and exit are fundamentally autonomous, reflecting the decentralized nature of these organizations. Unlike traditional organizations, where membership is often controlled by a central authority, DAOs allow individuals to join or leave at their own discretion, without requiring approval from existing members or higher authorities (Wright 2021). This autonomy in membership is facilitated by the DAO's use of tokens, which define organizational boundaries and grant members access to the DAO's resources and decision-making processes (Faqir-Rhazoui et al. 2021). The ability to enter and exit the organization independently without the approval of any formal control authority ('gatekeeper') is a key feature of DAOs, aligning them more closely with meta-organizations (Gulati et al. 2012) and online communities (Butler et al. 2014) than with traditional hierarchical entities.

Roles and responsibilities: informal

The roles and responsibilities within DAOs are typically informal and fluid, in contrast to the formalized roles found in traditional organizations. In DAOs, members often operate anonymously and independently, taking on tasks based on their interests and capabilities rather than being assigned specific roles by a central authority (Braun et al. 2021). This lack of formalized roles can lead to significant challenges in terms of accountability and responsibility, as there is often no clear delineation of who is responsible for specific actions or outcomes (Rikken et al. 2019; Braun et al. 2021). The decentralized and anonymous nature of DAOs can exacerbate these challenges, as members may be reluctant to take on accountability for their actions, knowing that their identities and actions are not easily traceable (Rikken et al. 2019). This informality in roles and responsibilities is both a strength and a weakness for DAOs: it allows for greater flexibility and autonomy, but it can also lead to governance issues, particularly in times of crisis when clear accountability is crucial (DuPont 2017; Beck et al. 2018).

Decision-making mechanism: decentralized

Decision-making in DAOs is decentralized and typically executed through on-chain voting mechanisms, where each token holder can participate in the decision-making process. This decentralized approach is a core principle of DAOs, distinguishing them from traditional organizations, where decision-making authority is often centralized in a hierarchical structure (Vergne 2020; Hsieh and Vergne 2023). Thus, decisions in a DAO would be made solely based on the blockchain-encoded procedures *on-chain* by the DAO token owners (Wang et al. 2019). However, based on practical experience with DAOs—most notably the often cited The DAO hack in 2017 (DuPont 2017)—this view of pure algorithmic governance in DAOs has not materialized, because reality is too complex to be codified perfectly. Particularly in emergency situations, quick actions are needed that are difficult to pre-implement fully on-chain. In the case of The DAO hack, the community decided off-chain to revert the state of the DAO to a point before the hack, thereby saving the funds of the DAO members. Thus, the prevailing view in the literature today is that DAOs clearly depend on the interaction and active participation of people also *off-chain*—i.e., outside the blockchain (e.g., El Faqir et al. 2020; Hassan and De Filippi 2021; Santana and Albareda 2022). A community of DAO members, which Hsieh et al. (2019) also refers to as *internal stakeholders* of the DAO, is thus fundamental to operate, manage, and develop the DAO through decentralized, transparent, and democratic consultation and decision-making processes (Wright 2021). In doing so, such decision-making processes can also be carried out through public social media channels, such as Discord, GitHub, Twitter or Reddit (Santana and Albareda 2022).

Organization's life cycle: decentralization

Early in their development, DAOs often exhibit centralized governance aspects, with only a small group of DAO founders taking key decisions (Ziolkowski et al. 2020). As they

mature, governance typically shifts towards decentralization, with decision-making power distributed more broadly among members, aided by smart contracts and on-chain mechanisms that reduce central control (Lumineau et al. 2021). However, DAOs may revert to centralization during crises to respond quickly, before returning to decentralized governance once stability is restored (DuPont 2017). This dynamic adaptability enables DAOs to remain relevant in a consistently evolving environment. In line with Mintzberg's (1984) life cycle model, many DAOs initially exhibit autocratic governance structures, which may be necessary for early development (Ziolkowski et al. 2020). Over time, they can transition to mission-driven or meritocratic governance, particularly in smaller, strongly aligned communities (Butler et al. 2014; Peña-Calvin et al. 2024). This shift is supported by the digitization of governance through smart contracts, which disempower traditional central actors by reducing uncertainties (Büchner 2018). DAOs can thus be seen as 'meta-organizations' with open boundaries and heterarchical decision-making (Gulati et al. 2012).

The middle section of Table 1 displays the analysis of DAOs through the lens of the previously developed *organizational design principles*, while highlighting the unique and innovative ways in which these organizations structure themselves to achieve their goals. From the shared goals established by a community of members to the decentralized decision-making mechanisms that define their governance, DAOs represent a significant departure from traditional organizational models. Their reliance on autonomous membership, informal roles, and incentive-based structures allows for a high degree of flexibility and adaptability, making them well-suited for the dynamic and often unpredictable environment of the digital age. However, these same features also present significant challenges, particularly in terms of accountability, decision-making speed, and governance stability.

Empirical analysis

Background

In the following, we briefly introduce three property-owning DAOs, before applying and analyzing our previously developed design principles for this type of DAO. The selection of the three examined DAOs—CityDAO, Cabin, and no1s1—was based on their distinct and illustrative contributions to the concept of property-owning DAOs. CityDAO was chosen as the first DAO to own physical assets, setting a precedent for the model. Cabin, with its well-documented and globally established framework, represents a DAO that integrates physical asset ownership within a broader community. No1s1, in contrast, provides a radical approach to self-ownership, where the property is owned by the organization itself rather than by individual token holders, offering a unique and extreme example of DAO ownership. Together, these DAOs provide a broad spectrum of ownership structures that enrich our understanding of property-owning DAOs in practice. It should be noted that the legal situation surrounding DAOs, including their ability to hold property, is an ongoing topic of research and debate. While some legal scholars, such as Pasquier and Bärtschi (2024), have explored the integration of DAOs within existing legal frameworks, this is not the focus of the current paper. The evolving nature of the legal status of DAOs and their implications within current regulations remain outside the scope of our analysis, but we acknowledge this ongoing discourse.

CityDAO

CityDAO was created in 2021 by a few enthusiasts driven by the vision of democratizing land ownership through a decentralized autonomous organization. Under the guidance of the core team, CityDAO was established as an LLC in Wyoming, issued its Citizen NFTs, and acquired a 40-acre plot of land (Crandall 2023). However, the core team still holds authority over CityDAO's $8.5 million treasury, with eight members acting as signatories on the organization's multi-signature wallet. Following the land purchase in Wyoming, CityDAO members deliberated on its utilization. Eventually, in November 2021, the community endorsed a proposal designating Parcel 0 primarily for conservation and wildlife purposes. This decision prompted CityDAO to contemplate the future direction of the organization, as Parcel 0 was earmarked for conservation rather than the envisioned construction of a "city of the future" on the blockchain. CityDAO provided an app for the community to view different plots of Parcel 0 featuring the specific properties, e.g., high avalanche risk, or badlands. Users can also view the ecological composition describing the soil and the vegetation of a specific plot. The app allows landowners to interact with and make informed decisions on the land.

The governance of CityDAO subsequently evolved through several iterations (Rong and Mao 2023): first with a core team, then with a council, followed by guilds, and finally with a project-based model. Each phase reflected the tensions between centralization and decentralization, efficiency and effectiveness. The establishment of the council marked a shift towards decentralization and culminated in the ratification of a charter. However, the guild structure faced challenges, such as resource silos and inefficiencies, which led to a shift to project-based governance. Despite criticism of accountability and resource allocation, projects such as the Baby Parcel Initiative in Colorado were launched. CityDAO currently faces

a severe governance dilemma, with members debating future direction and struggling with operational disruptions. As a result of these discussions, CityDAO decided to re-focus its operations and become active in the real-estate market allowing token holders to participate in property transactions, rentals, lodging, and futures trading. It bridges the physical and digital world by tokenizing physical properties. While CityDAO does not anymore manage land directly, token holders may still participate in the governance by voting on the platform's development and operation.

Cabin

Cabin was founded with the vision of creating a network city, where people from all over the world come together to live and collaborate in the middle of nature (Cabin 2024). At the heart of Cabin is the expectation that future urban landscapes will adopt an online organizational model and decentralize the physical presence in different locations. The foundation of Cabin's collective framework is based on polycentric governance that includes neighborhoods, guilds, communities, and citizens, each taking on different roles within the community. Depending on their role and contributions, citizens are given different responsibilities and voting rights.

Cabin strives to provide housing alternatives that are more economical and of higher quality while also being more adaptable than traditional urban environments. It does not strive to make profits or to distribute dividends. Citizens may buy 'passports' that facilitate access to community housing and apartments in Cabin neighborhoods. Cabin's online community has helped create the first housing community, Neighborhood Zero, in Texas' Hill Country. Neighborhood Zero includes a four-bed room cabin on a plot of land and it is sustained by a digital community: four guilds manage the digital, physical, business, and social aspect of the DAO. Neighborhoods can thereby publish their offerings and needs to global citizens via an online marketplace. This collaborative ecosystem allows citizens to either pay for their stay or contribute their expertise in exchange for housing or payments. Members of the DAO can complete bounties (tasks) to obtain tokens that they can use to vote inhouse or to sell them on a marketplace. They have also organized building weeks, where members met physically to build another cabin. As of the time of writing, there are 13 neighborhoods across the world (USA, Europe, and Asia). Neighborhoods can be created bottom-up by members enabling an organic growth as they are built wherever they are needed. New neighborhoods can apply for funds from the DAO to support their creation and to be added as a new physical node in Cabin network city. Through this, Cabin connects a digital community with physical places that are governed by a DAO with its own token.

no1s1

In 2018, the Swiss think tank *Dezentrum* introduced the concept of a self-owned and self-managed house with no1s1 (*no one's one*). They termed it a 'future experiment' ('Zukunftsexperiment') aimed at exploring novel governance approaches to collective goods facilitated by blockchain. While conventional experiments probe existing social realities, future experiments probe social possibilities. However, future experiments are not merely speculative but aim to form the future. Consequently, in subsequent years, an initial prototype of no1s1 was developed at ETH Zurich (Hunhevicz et al. 2021; Hunhevicz 2022), primarily focusing on the technical feasibility of a blockchain-based autonomous space. To better investigate the concept of a DAO that owns properties, a research partnership was formed between the University of Zurich and ZHAW Zurich University of Applied Sciences. The project is funded by the Digitalization Initiative of the Zurich Higher Education Institutions (DIZH). The goal of no1s1 is to enable the self-management of a physical space. For this, the prototype features an architecture connecting the digital and physical world (Hunhevicz et al. 2021). The digital aspect consists of the blockchain part, with smart contracts managing funds, access, and potentially providing the layer for the governance framework, even though this was not implemented. Spychiger et al. (2024) provide a description and simulation of a governance framework for no1s1 featuring a use-based membership. Users need to interact with the physical house first to obtain membership. A web-based frontend allows users to digitally book the space in advance.

The physical aspect consists of the house itself (the physical shell) and several IoT-devices that steer and monitor the house's functionalities, but also allow for physical interactions with the users. Therefore, no1s1's architecture features a digital decision-making process on physical assets that the users can directly interact with. The house can also get signals from the physical infrastructure to trigger decisions and actions, e.g., maintenance tasks. This connection of the physical space with the digital space allows the house to interact with its environment and autonomously start to self-manage its own use and maintenance. The possible advantage of such an approach are more efficient processes, lower fees, and since no1s1 is a DAO, a more inclusive decision-making approach.

As no1s1 belongs to no one but itself, no single entity can decide its fate. However, no1s1 interacts with its users. While the primary purpose of no1s1 is clearly to sustain its existence, its usage might vary over time. To cope with the changing needs of its users and its environment, no1s1

needs to remain adaptable. This needs proper organizational structures and governance mechanisms to define the community's and individuals' rights, obligations, and responsibilities, as well as the related processes. For this, tokens which reflect specific entitlements of the users, for example, membership, can be issued. Furthermore, users may book the house and pay for the use of the physical space of no1s1 with cryptocurrencies. No1s1 can then use this income to pay contractors for maintenance activities. In addition, the tokens can be earned by actually using the house and enable users to participate in the governance of no1s1.

Application of the DAO design principles to property-owning DAOs

Using the previously established DAO design principles (Table 1), the three property-owning DAOs are evaluated along the core organizational design principles of *purpose, membership,* and *governance* and its six sub-categories.

Shared goal (purpose)

The purpose of property-owning DAOs is to provide a living and working space for a broader community. To this end, they need to own a property itself and manage its maintenance and use by the community. A property is a limited and reality-bound, physical object. This is a completely different situation to digital assets, which are in principle always expandable, divisible, and shapeable. We see in CityDAO that property can initially be just a piece of land, whereas no1s1 and Cabin already are built around a living space. However, CityDAO has also developed more toward physical buildings, as a piece of land is not very useful without any infrastructure on it. As Cabin clearly highlights, there should be no profit distribution sought; rather, the focus is on community values for the shared creation of a community. In addition, no1s1 aims to enable more efficient management of infrastructure governed by the community. In doing so, property-owning DAOs are guided by a shared mission that is strongly accepted by all organizational members. The profit generated by the use of the property automatically returns to the treasury, which can be again used by the community for operating costs, such as maintenance (no1s1), growth (Cabin), or reinvestment (CityDAO). Consequently, the longevity of property-owning DAOs is highly dependent on the usage and individual contributions of the community members, creating values they are aligned with.

Incentive structure (purpose)

Property-owning DAOs such as no1s1, CityDAO or Cabin own and maintain a physical infrastructure, such as a house or piece of land, with users incentivized to contribute to the well-being of the property by performing specific tasks and paying for the usage. The shared purpose within the community and a properly aligned incentive structure should thereby increase community participation, property usage, and maintenance activities. To implement its incentive system, property-owning DAOs use various tokens that promote specific tasks performed by the community. As Cabin highlights, the token system should place a clear focus on the use and the maintenance of the living space and the community, with different stakeholders incentivized to contribute to the purpose of the DAO through different actions—be it in the digital (writing posts, translating webpages) or in the physical world (building houses). The prices for the different tokens and thus the price for the use of a property could be determined by the community (for no1s1 see Spychiger et al. 2024) or also coded in smart contracts (e.g., fixed price calculation depending on maintenance costs) or based on oracles (e.g., the prices for the use of properties could be automatically taken from conventional rental websites). It is important to note that the focus is on optimizing the use of an inherently limited resource, which aligns with a sharing logic aimed at distributing what already exists rather than the market and growth logic that seeks overall efficiency gains. Hence, this approach stands in contrast with the traditional concept of property, underscoring the transformative potential of digital technologies in reshaping how commons-based resources are produced and shared. While growth, such as acquiring or creating additional property is possible in CityDAO and Cabin, it is viewed as an option with both advantages and disadvantages for members and can still be rejected by the community. This consideration is closely linked to the effective organization of membership and governance.

Autonomous entry and exit (membership)

The members of a property-owning DAO are the users of the property, who also manage and define the state of the living space through processes that are kept open and transparent. In this context, membership is coupled to the use of the living space as with the passports in Cabin. Even more so, in no1s1, interactions with the physical property constitute a kind of membership token with membership rights. However, in property-owning DAOs, there is an upper capacity limit to membership, since the physical space is limited. One possibility to cope with this is a pricing model: to use the living space, interested community members would need to book the space in advance and pay according to a

community-defined dynamic pricing model. With the aim of responding to an imbalance between supply and demand, a smart contract-based dynamic pricing system would give the DAO more flexibility to manage demand. However, at one point the usage cost might be too high and shared usage is not possible anymore, defeating the purpose of the DAO. In this case, the DAO could expand its physical space, such as in Cabin, where members can create new neighborhoods. Such an organic growth would then again create room for more members that use the space. This is important, since if the membership is not coupled to the usage—as we can observe with the Citizen NFTs in CityDAO—the online community can quickly grow and become detached from the physical living space itself and the general interest in maintaining its usability for the community can give way to a purely speculative motivation for financial reasons. Therefore, leaving the property-owning DAO and losing membership rights corresponds with not using the living space anymore; hence the membership token will lose value over time. In Cabin, passports are time-limited to implement autonomous membership based on usage. The membership and organizational boundaries of the DAO are, therefore, clearly focused on the purpose of 'shared property usage', while the accumulation of membership power or token value is hardly possible to avoid centralization.

Informal roles and lack of accountability (membership)

The DAO members use the living space autonomously but decide on its future developments jointly. In Cabin, users can autonomously create new neighborhoods, but need the approval of the whole community for its funding. All three DAOs have the possibility to interact with third parties if needed, so that contributions can be commissioned externally. It is the task of a property-owning DAO to maintain a community-driven living space for its members, provide a range of services, and motivate members to take part in organizational processes. For this, Cabin and no1s1 provide incentives to fulfil some tasks and any member can participate—no formal roles exist. However, as members of the DAO come into physical contact with the living space by fulfilling tasks or using the space, they lose their anonymity to some extent, and they can be held responsible, at least to some degree, for bad behavior within the community. Incentivizing to do the right thing and linking one's decisions to one's surroundings offers, therefore, at least a limited accountability within property-owning DAOs.

Decentralized decision-making (governance)

As discussions in CityDAO indicate, it is critical to guarantee that members have enough incentives and interest to participate in decision-making processes for property-owning DAOs. However, it is equally important that maintenance issues related to the properties will be resolved in a self-executing fashion and that potentially important work will not go undone. For example, no1s1's governance processes target the well-being for no1s1 and an active participation by the community. To guarantee the proper continuity of the digital and physical elements of the DAO, and become a viable organization, it is the goal of Cabin to have tasks resolved by independent actors incentivized through bounties. The self-managing maintenance can be seen as the collective administration of property-owning DAOs, comparable to a co-shared facility management providing certain services for its properties. Furthermore, in no1s1 a house can govern itself through a series of self-executing mechanisms enabled by IoT-devices that help it to sense its environment. While the community or even the property itself can take over most of the day-to-day administration, some larger decisions require a community decision-making process, where uncertainties will be reduced through predefined on-chain rules (smart contracts). For example, the funding of a new neighborhood in Cabin or the designation of Parcel 0 as wildlife conservation area in CityDAO needed a community-wide vote.

Decentralization life cycle (governance)

Property-owning DAOs can be seen as an evolution of power structure—from private-ownership to an on-chain community of property users—using the features of the blockchain to coordinate decision-making among its members. The formation of these projects is usually done relatively centralized through a private founding team (CityDAO, Cabin) or a publicly funded research project (no1s1). The ways how such a property then gets transitioned into the ownership of a DAO can be manifold: (i) it could be donated by someone (Cabin); (ii) it could be handed over to the DAO by a private or public institution for a fixed time period to conduct experimental research (no1s1); or (iii) similar to CityDAO, the community donates money to the DAO to buy the property. Over time, different structures may emerge, such as the guilds in CityDAO or the neighborhoods in Cabin. Developments in property-owning DAOs are strongly linked to the members themselves, who initiate participatory change of their living environment, establish discussions among themselves and maintain an open governance process. If the community is not interested in using the property, the distributed tokens expire, and the DAO self-liquidates. A constant demand in wanting to use the property hereby ensures the existence of the property-owning DAO.

Conceptualization of DAO design principles for property-owning DAOs

Building on the three property-owning DAO projects and the application of the DAO design principles, we can structure the different organizational requirements to further develop the organizational design of a physical space governed by a DAO. The application of the DAO Design Principles to the three cases—no1s1, CityDAO, and Cabin—has shown that, for property-owning DAOs, physical space and its usage are a predominant design feature. A shared property usage defines the purpose, which is also incentivized by the DAOs. Furthermore, the property is instrumental in determining membership and its associated tasks. Regarding governance, the property is the subject of governance and is self-administered by the community. Eventually, along the life cycle, power structures may evolve around the property, and its active usage is decisive for the survival of the DAO. Therefore, a consolidation of the organizational principles via the principles for DAO in general into the resulting principles of DAO for property-owning DAOs is synthesized in Table 1.

Discussion

This study has indicated that there are six design principles of a traditional organization that are also relevant for the design of DAOs. With property-owning DAOs, we conceptually made use of a specific type of DAO, namely, a DAO with physical living space in the form of houses or land. In an initial line of argument, we can note that DAOs generally have a purpose that is meant to benefit a community. By putting community benefits at their center, DAOs can create a unique value proposition for all their stakeholders that more traditional ownership-based businesses often fail to deliver. A DAO is, therefore, highly dependent on a strong community-wide dedication to the purpose, otherwise conflicts of interest and political governance discussions could arise (DuPont 2017; Beck et al. 2018). This is because in a decentralized organization, there are no hierarchical structures that could resolve and eliminate conflicting uncertainties in timely manner. For the governance of a DAO, therefore, a sustained purpose accepted by all, in the sense of Mintzberg's (1983) *ideology* and *mission*, is vital. This has also been shown in research on more open and community-based organizations, such as meta-organizations (Gulati et al. 2012) or online communities (Butler et al. 2014).

To create and maintain such a commonly accepted ideology as a governing structure in a DAO over time, it is important to develop and implement the right incentive structure (Faqir-Rhazoui et al. 2021), and organizational membership should be linked to active use of the DAO to limit uncontrolled growth (Peña-Calvin et al. 2024). This serves to keep the current intentions of the members aligned, to exclude inactive members and, finally, to attract only new members that share and strengthen the guiding mission of the DAO. This is even more important in a DAO than in a traditional organization, because DAOs generally have fairly open and autonomous entry and exit rules for members (Braun et al. 2021). In a decentralized organization there should be no entity acting as gatekeeper defining the organizational boundaries. For property-owning DAOs we have proposed a membership mechanism that is related to the purpose of the DAO. It is in this way we believe there is a chance that new and old DAO members, who cannot be assessed for their intentions and prevented from joining or leaving the organization by any gatekeeper, will share the common mission of the DAO and contribute to the well-being of the property-owning DAO.

However, it is questionable if a decentralized organization can hold autonomous—and maybe even anonymous—members accountable for their interactions with the DAO (Braun et al. 2021). In this context, DAOs face the difficult tension between the establishment of formalized rules to structure certain organizational processes like decision-making and the subsequent non-formalized and undefined execution of the outcome of such processes by their autonomous and often unaccountable actors (Kühl 2021). Not surprisingly, many DAOs tend to choose centralized off-chain mechanisms for their governance processes, especially at the beginning of the development and during the organization's expansion, with the intention of implementing on-chain mechanisms later in the organization's life cycle and finally reaching decentralization (Ziolkowski et al. 2020). With respect to property-owning DAOs, an example to this conflict of off-chain and on-chain governance can be seen in CityDAO. Governed by a core team in the beginning, they have over the years decentralized decision-making and are now mainly organized to different project teams (Rong and Mao 2023). However, to keep teams and its members accountable for their actions and decisions is difficult. By now, they are experiencing political conflicts, as members no longer agree on the purpose and further development of the DAO. It has been even proposed to liquidate CityDAO.

Looking at the variety of different tasks in property-owning DAOs, it seems likely that some sort of organizational structure with members having different roles, duties and rights will emerge (Luhmann 2000). Consequently, some sort of hierarchies will be implemented as now not all the members are equal anymore having the same rights and obligations. Nevertheless, as also outlined by Vergne (2020), hierarchies may emerge, but an organization can still be decentralized. In contrast to traditional organizations the determination and the delegation

of rights and duties as well as the performance control should be governed bottom-up by the community in an open, decentralized, and transparent way based on encoded on-chain smart contracts (Murray et al. 2021). Hereby, transparency seems to be a very important—but sometimes underestimated—design principle of DAOs. As there is no central control authority in DAOs, it is crucial that the community has full access to all information and processes to comply with the important role as decentralized control entity. Only if all members have full access to all relevant information, an organization can be considered decentralized in its governance, as otherwise some members will always benefit from their information advantage and influence the decision-making process for their own purpose. Furthermore, over the life cycle of the DAO it is important that the members can initiate a procedure to adapt the current processes to the changing requirements of the community and the environment. This is important insofar as members might be kept legally accountable for decisions made in the DAO (Riva 2021). It is, therefore, questionable whether the anonymity of its members is a viable option for all DAOs—and not only from a legal but also from an organizational and technical point of view (Rikken et al. 2019).

As compared to blockchain governance of purely on-chain organizations (Vergne 2020; Lumineau et al. 2021; Hsieh and Vergne 2023), property-owning DAOs could leverage the physical interaction of its members when they enter the living space to address the anonymity issue. This physical presence might be a valid moment to create a unique digital identification for each member and to establish some limited accountability for each of them. Compared to many DAOs, property-owning DAOs' physical presence and the need for its members to interact with the living space in a physical way is a fairly unique feature. This physical manifestation entails three important considerations compared to more traditional DAOs with purely virtual existence (e.g., decentralized finance): *first*, by interacting with the property, members could personally meet each other off-chain in the real world, which could strengthen the sense of a community; *second*, in the event of disagreements within the community, the split (*fork*) of the DAO would not be a viable option for resolving the conflict due to the indivisibility of the property, which may create a stronger desire to agreements and more system stability; and finally *third*, because the physical interaction is constrained to a geographic space, property-owning DAOs will have stronger roots in the local community, which could lead to tighter ties and identification among its members. Therefore, these physical aspects affect the decision-making processes and the autonomy of these DAOs defining the organization in the sense of Luhmann (2000).

However, the physical presence might also create some new issues as global scaling is much more difficult. Scaling can thereby be considered critical as decentralization in DAOs heavily relies on the participation of a broad and diverse community. To avoid this matter, the properties must be of use to a sufficiently large number of people, otherwise, it would be difficult to gather enough members for a decentralized organization and off-chain agreements between a few local members would become more likely. Having only one property with a purposeful use for a limited number of members would therefore, not justify the establishment of a DAO. This could explain the difficulties in governing CityDAO, or that no1s1 with only one living space (house) can only be an experimental research environment. Similar to Cabin, property-owning DAOs may, therefore, only be relevant if they govern multiple properties that are used by a broad, geographically dispersed community for occasional use—e.g., for vacations, seminars, meetings, events, or workshops.

Research agenda

From an organizational theory standpoint, research has begun assessing the impact of blockchain technology on traditional organizations, suggesting that it could decentralize communication and distribute decision-making (Vergne 2020), thereby enabling cooperation and coordination within organizations without the need for a central authority (Lumineau et al. 2021; Hsieh and Vergne 2023). However, as Lumineau et al. (2021) highlight in their comprehensive research agenda, blockchain technology presents numerous opportunities for organizational scholars to investigate its potential in organizing collaboration. Nonetheless, current literature predominantly focuses on digital platforms, open-source software communities, and virtual cryptocurrencies. In this context, our conceptual research offers valuable insights into the potential organizational structures of property-owning DAOs, contributing a significant new perspective to the ongoing discussion about the impact of blockchain on organizations.

Despite its unique research focus, our study leaves many questions unanswered for future research. In this study, we did not take legal and property aspects into consideration. Of course, the question of the legal entity that would frame the organization is important, but also complex, and depending on the country and the exact purpose and goals of the DAO, the answer would certainly vary tremendously. The legal position of DAOs is currently not well defined and often discussed by scholars and practitioners in blockchain research (e.g., De Filippi and Wright 2018; Riva 2021; Schuppli and Jafari 2021; Pasquier and Bärtschi 2024). An example of a new law and a new form of legal entity for blockchain-based

organizations has been found in the US state of Wyoming, which has recognized and regulated DAOs as limited liability companies (DAO LLCs) since July 2021.[5] CityDAO is constituted on the base of this law. The meanwhile revised law concerns the rules on entry and exit, as well as the clarification that DAOs are neither purely member managed nor strictly algorithmically managed but may tend towards one of these types of governance depending on the specific design. However, the legal discussion might provide a clearer idea what a DAO is and how it should be structured and implemented to comply with regulations.

Under this viewpoint, we have also not touched the question how a property could become its own owner so it can be called *self-owned* or how a property like a house can be integrated into a DAO in a way that not the DAO members are automatically the owners of the property. In our view, the intention of property-owning DAOs is not to create a co-ownership of a private property, but rather a *co-usage* of a common property. How such co-usage of a common property needs to be managed has been widely studied for environmental resources managed by communities (e.g., Ostrom 1990) and in the collaborative housing research (e.g., Lang et al. 2020). For example, rural commons where communities manage shared resources, or housing cooperatives, where members collaboratively maintain and govern properties, reflect similar governance structures to those seen in property-owning DAOs (e.g., Van Vulpen and Jansen 2023; Lustenberger et al. 2024). These real-world models provide valuable insights into the management of shared resources and the governance of common property. Therefore, it would be interesting to discuss how the community-based approaches to common-pool resources and collaborative housing management could benefit from the new technological possibility of DAO governance—and vice versa, as such research might also provide valuable insights into how the governance of property-owning DAOs such as no1s1 or Cabin needs to be structured to avoid free-riding issues, ownership appropriation (centralization) and deterioration of the common property so that the long-term viability of the DAO is assured.

As Faqir-Rhazoui et al. (2021) state, an important aspect of decentralized governance of a DAO is an appropriate incentive structure to incent the different management tasks that need to be fulfilled for the DAO. As DAOs currently do not have a self-owned intelligence that acts on their behalf, they are heavily dependent on DAO members and the broader community for decision-making and performing specific actions. That community may provide this intelligence in the sense of the 'wisdom of the crowd' with the help of oracles for the DAO. Especially in property-owning DAOs, where properties should be *self-managed* by a *self-governed* DAO, an active participation of the DAO community is vital. The members of this community are autonomous in their decisions and actions, which implies that they do not have any obligations to participate nor to act. Therefore, it is critical to incentivize all administrative and maintenance tasks in a sufficiently motivating manner so that individuals take on these tasks and perform them correctly. A further solution could be to think about incentives in the form of reputation to uphold the DAOs requirements (Spychiger et al. 2022). Members could earn reputation for positive actions (e.g., leaving the property on time and in good conditions) and again lose it for passivity or negative actions (damaging the property). These incentives can be implemented through a reputation token that could become even more attractive in combination with economic incentives or utility, such as a salary or some payments in relation to the reputation stake (Kaal 2021). However, the design of such token systems is a field that deserves more research.

With the development and evaluation of technologies, new design opportunities for DAOs will emerge in the future. In addition to innovations in the blockchain technology, another interesting technology related to DAOs is artificial intelligence (e.g., Salah et al. 2019; Shrestha et al. 2019). In terms of property-owning DAOs, people may only interact with the property as customers to use it or as contractors to be paid to perform certain activities. People will no longer be members of the DAO, as all decisions could be taken by an artificial intelligence based on predefined smart contracts running on a blockchain. A scenario certainly worth thinking about—especially if we imagine that such an artificially intelligent DAO could be our future landlord.

In our work, we consider the traditional strands of organizational theory mainly form the systems theory perspective of Luhmann (2000). For sure there are other organizational theory perspectives that can be relevant in looking at DAOs, most notable the network theory perspective (Jones et al. 1997; Powell 1990; Gulati et al. 2012) with its focus on hierarchy-less, spontaneous and temporary organizational structures and boundaries. Other aspects of governance could also be included, such as organizational performance and its control, and the measures derived from it to maintain or enhance organizational performance (Hoverstadt 2008). By integrating a systemic cycle of continuous performance measurement and organizational adaption, a closer examination of more complex concepts of organizational design, such as the viable system model (Beer 1985; Espejo 1989; Schwaninger 2009) could additionally provide a broader understanding of possible organizational models for DAOs in general and property-owning DAOs specifically.

[5] https://wyoleg.gov/2022/Enroll/SF0068.pdf.

Conclusion

Property-owning DAOs are defined as a combination of physical living spaces and virtual decision-making communities. With regard to the different concepts and experiments already initiated, such as Cabin, CityDAO and no1s1, the design of the organizational structures and the long-term viability of property-owning DAOs is not yet clear. While the implementation of a physical infrastructure regulated by booking and payment transactions via the blockchain is possible, virtual organizational properties and the design of functioning community structures around the concepts of property-owning DAOs have not received any attention so far. Driven by the interconnection of the tangible and intangible, as well as the interplay of physical and digital assets, new organizational structures are emerging that are changing the current understanding of organizations and opening new directions in organizational research. This research paper provides insights into the organizational design of new decentralized organizations that manage a physical property through strict program logics and open decision-making processes, as well as autonomous memberships. While the connection between decentralized autonomous organizations and properties forms a nascent area of organizational research, it is not the goal of this article to map a comprehensive representation of all possible organizational models at this interface.

This study examined the organizational system theory (Luhmann 2000; Mintzberg 1983) and synthesized the findings with the DAO literature. The possibilities for designing DAOs are manifold, and our research was able to identify three main key organizational principles of *purpose*, *membership*, and *governance*. Subsequently, we proposed six *DAO Design Principles* based on the previous understanding of decentralized autonomous organizations and the linkage to the organizational system theory literature. To finally answer the research question, we then applied these general *DAO Design Principles* to property-owning DAOs.

The design of the organizational structures of property-owning DAOs is characterized by its clear shared purpose to provide *shared property usage* for a broader community, whereby the community gets incentivized to contribute to the well-being of the shared property. Membership in property-owning DAOs is closely tied to ongoing interactions with the property and, therefore, depends on active participation in DAO's organizational activities rather than a one-time investment in an asset. Within property-owning DAOs the members will need to collectively govern all activities as a community, whereby these self-governing aspects are structured through a series of predefined and self-executing mechanisms encoded on a blockchain. The community of property users must be able to change these coded self-governing rules over the life cycle of the DAO to meet the demands of an ever-changing environment and remain adaptable as an organization.

To further explore the connection of decentralized autonomous organizations and properties, analyzing further existing DAOs that experiment with properties on the base of our develop framework is a suitable next research approach. However, as already highlighted in the research agenda, there are still many open research questions in respect to DAOs (Tan et al. 2023). The implication from further theoretical as well as applied research in the field of legal studies, governance of communities and modern organizational theories is yet to be understood. Our conceptual research can be seen as a first attempt to get more insight into the new phenomenon of decentralized autonomous organizations owning properties. Our research around decentralized management of living space has a high degree of novelty in terms of theory and practice which can further advance the academic discussions in the domain of collaborative housing research and add to the discussions in the context of facility management, exploring new ways of maintenance systems. Our conceptual findings contribute to these studies and can be used as a starting point to further develop and enhance real-world implementations.

Author contribution All authors contributed to the study's conceptualization, design, and execution. Michael Lustenberger managed the project and together with Florian Spychiger led the conceptualization and methodology. Material preparation, literature review, and drafting were conducted by Michael Lustenberger, Florian Spychiger, and Lukas Küng. Jens Martignoni provided feedback on the manuscript and assisted with the systems theory perspective. All authors contributed significantly to reviewing and editing and approved the final manuscript.

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Data availability Not applicable.

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References

Atzori M (2015) Blockchain technology and decentralized governance: is the state still necessary? SSRN Journal. https://doi.org/10.2139/ssrn.2709713

Beck R, Müller-Bloch C, King J (2018) Governance in the blockchain economy: a framework and research agenda. J Assoc Inf Syst 19:1020–1034

Beer S (1985) Diagnosing the system for organisation. Wiley, New York

Bevir M (2012) Governance: a very short Introduction, 1st edn. University Press, Oxford

Braun A, Häusle N, Karpischek S (2021) Incentivization in decentralized autonomous organizations. SSRN J. https://doi.org/10.2139/ssrn.3760531

Büchner S (2018) Zum Verhältnis von Digitalisierung und organisation: on the relationship of digitization and organization. Z Soziol 47:332–348

Buterin V (2013) A next generation smart contract & decentralized application platform. Whitepaper Ethereum Foundation. https://ethereum.org/en/whitepaper/. Accessed 11 July 2023

Buterin V (2014) DAOs, DACs, DAs and more: an incomplete terminology guide. ethereum foundation blog. https://blog.ethereum.org/2014/05/06/daos-dacs-das-and-more-an-incomplete-terminology-guide/. Accessed 10 May 2023

Butler BS, Bateman PJ, Gray PH, Diamant EI (2014) An attraction-selection-attrition theory of online community size and resilience. MIS Q 38(3):699–728

Cabin (2024) Building Cabin's Network City. https://creators.mirror.xyz/zHox-AfvHbX0Q2JxAbcfwiAmOXRy7BzYafhZNlvHnmM. Accessed 12 May 2024

Crandall J (2023) Living on the block: how equitable is tokenized equity? Big Data Soc. https://doi.org/10.1177/20539517231208455

De Filippi P, Wright A (2018) Blockchain and the law: the rule of code. Harvard University Press, Cambridge

DuPont Q (2017) Experiments in algorithmic governance. Bitcoin and beyond. Routledge, London, pp 157–177. https://doi.org/10.4324/9781315211909-8

El Faqir Y, Arroyo J, Hassan S (2020) An overview of decentralized autonomous organizations on the blockchain. In: Proceedings of the 16th international symposium on open collaboration. Presented at the OpenSym 2020: 16th international symposium on open collaboration, ACM, Virtual conference Spain, pp 1–8. https://doi.org/10.1145/3412569.3412579

Espejo R (1989) A method to study organizations. In: Espejo R, Harnden R (eds) The viable system model: interpretations and applications of Stafford Beer's VSM. Wiley, Chichester, pp 361–382

Faqir-Rhazoui Y, Arroyo J, Hassan S (2021) A comparative analysis of the platforms for decentralized autonomous organizations in the Ethereum blockchain. J Internet Serv Appl 12(1):1–20. https://doi.org/10.1186/s13174-021-00139-6

Feichtinger R, Fritsch R, Vonlanthen Y, Wattenhofer R (2024) The hidden shortcomings of (D)AOs—an empirical study of on-chain governance. In: Essex A, editor. Financial cryptography and data security. FC 2023 international workshops. FC 2023. Lecture notes in computer science, vol 13953. Springer, Cham. https://doi.org/10.1007/978-3-031-48806-1_11.

Fulmer IS (2012) Editor's comments: the craft of writing theory articles—variety and similarity in AMR. Acad Manag Rev 37(3):327–331

Gilson LL, Goldberg CB (2015) Editors' comment: so, what is a conceptual paper? Group Org Manag 40(2):127–130

Gioia DA, Pitre E (1990) Multiparadigm perspectives on theory building. Acad Manag Rev 15(4):584–602. https://doi.org/10.2307/258683

Goldsby C, Hanisch M (2022) The boon and bane of blockchain: getting the governance right. Calif Manage Rev 64(3):141–168. https://doi.org/10.1177/00081256221080747

Gulati R, Puranam P, Tushman M (2012) Meta-organization design: rethinking design in interorganizational and community contexts. Strateg Manag J 33(6):571–586. https://doi.org/10.1002/smj.1975

Hassan S, De Filippi P (2021) Decentralized autonomous organization. Internet Policy Rev 10(2):1–10. https://doi.org/10.14763/2021.2.1556

Heo K, Yi S (2023) (De)centralization in the governance of blockchain systems: cryptocurrency cases. J Organ des. https://doi.org/10.1007/s41469-023-00138-w

Hoverstadt P (2008) The fractal organization: creating sustainable organizations with the viable system model. John Wiley & Sons, West Sussex

Hsieh YY, Vergne JP (2023) The future of the web? The coordination and early-stage growth of decentralized platforms. Strateg Manag J 44(3):829–857. https://doi.org/10.1002/smj.3455

Hsieh YY, Vergne JP, Anderson P, Lakhani K, Reitzig M (2019) Correction to: Bitcoin and the rise of decentralized autonomous organizations. J Organ des 7(1):1–16. https://doi.org/10.1186/s41469-019-0041-1

Hufty M (2011) Investigating policy processes: the governance analytical framework (GAF). In: Wiesmann UM, Hurni H (eds) Research for sustainable development 6. Geographica Bernensia, Bern, pp 403–424. https://doi.org/10.7892/BORIS.68343

Hunhevicz JJ, Wang H, Hess L, Hall D (2021) No1s1 – a blockchain-based dao prototype for autonomous space. In: proceedings of the 2021 european Conference on Computing in Construction, vol 2. University College Dublin, pp 27–33. https://doi.org/10.3929/ETHZ-B-000504485

Hunhevicz JJ (2022) Exploring the potential of blockchain and cryptoeconomics for the construction industry. Doctoral dissertation, ETH Zurich

Jaakkola E (2020) Designing conceptual articles: four approaches. AMS Rev 10:18–26. https://doi.org/10.1007/s13162-020-00161-0

Jones C, Hesterly W, Borgatti S (1997) A general theory of network governance: exchange conditions and social mechanisms. Acad Manag Rev 22(4):911–945

Kaal WA (2021) A decentralized autonomous organization (DAO) of DAOs. SSRN J. https://doi.org/10.2139/ssrn.3799320

Kühl S (2020) Organisationen—was ist das eigentlich? Organisationen. VS Verlag für Sozialwissenschaften, Wiesbaden, pp 9–22. https://doi.org/10.1007/978-3-531-93185-2_1

Kühl S (2021) Organizations: a short introduction. Organizational Dialogue Press, Princeton

Lang R, Carriou C, Czischke D (2020) Collaborative housing research (1990–2017): a systematic review and thematic analysis of the field. Hous Theory Soc 37(1):10–39

Larimer S (2013) Bitcoin and the three laws of robotics. Let's talk bitcoin. https://letstalkbitcoin.com/blog/post/bitcoin-and-the-three-laws-of-robotics. Accessed 13 Apr 2022

Luhmann N (2000) Organisation und Entscheidung. VS Verlag für Sozialwissenschaften, Wiesbaden. https://doi.org/10.1007/978-3-322-97093-0

Lumineau F, Wang W, Schilke O (2021) Blockchain governance—a new way of organizing collaborations? Organ Sci 32(2):500–521. https://doi.org/10.1287/orsc.2020.1379

Lustenberger M, Wollenschläger S, Küng L (2024) DAO as digital governance tool for collaborative housing. SSRN J. https://doi.org/10.2139/ssrn.4950821

Meredith J (1993) Theory building through conceptual methods. Int J Oper Prod Manag 13(5):3–11
Miebach B (2007) Organisationstheorie: problemstellung—modelle—Entwicklung. VS Verlag für Sozialwissenschaften, Wiesbaden. https://doi.org/10.1007/978-3-531-90411-5
Mintzberg H (1983) Power in and around organizations. Prentice-Hall, Englewood Cliffs
Mintzberg H (1984) Power and organization life cycles. Acad Manag Rev 9(2):207–224. https://doi.org/10.2307/258435
Mowshowitz A (1997) Virtual organization. Commun ACM 40(9):30–37. https://doi.org/10.1145/260750.260759
Murray A, Kuban S, Josefy M, Anderson J (2021) Contracting in the smart era: the implications of blockchain and decentralized autonomous organizations for contracting and corporate governance. Acad Manag Perspect 35:622–641. https://doi.org/10.5465/amp.2018.0066
Ostrom E (1990) Governing the commons: the evolution of institutions for collective action, 1st edn. Cambridge University Press, Cambridge. https://doi.org/10.1017/CBO9780511807763
Pasquier B, Bärtschi H (2024) Dezentrale autonome Organisationen für Immobilien. Gesellschafts- und Kapitalmarktrecht 2024:200–217
Peña-Calvin A, Arroyo J, Schwartz A, Hassan S (2024) Concentration of power and participation in online governance: the ecosystem of decentralized autonomous organizations. In: Companion proceedings of the ACM on web conference 2024, pp 927–930. https://doi.org/10.1145/3589335.3651481
Powell WW (1990) Neither market nor hierarchy. Res Org Behav 12:295–336
Reijers W, Wuisman I, Mannan M, De Filippi P, Wray C, Rae-Looi V, Cubillos Vélez A, Orgad L (2021) Now the code runs itself: on-chain and off-chain governance of blockchain technologies. Topoi 40:821–831. https://doi.org/10.1007/s11245-018-9626-5
Rikken O, Janssen M, Kwee Z (2019) Governance challenges of blockchain and decentralized autonomous organizations. Information Polity 24:397–417. https://doi.org/10.3233/IP-190154
Riva S (2021) Decentralized Autonomous Organizations (DAOs) in the Swiss Legal Order. In: Bonomi A, Romano GP (eds) Yearbook of private international law, vol. XXI - 2019/2020. Verlag Dr. Otto Schmidt, Köln, pp 601–638. https://doi.org/10.9785/9783504386962-028
Rong H, Mao Z (2023) Deep-dive Into CityDAO: an experiment in collective land ownership and decentralized governance. Harvard Kennedy School, Cambridge
Salah K, Rehman MHU, Nizamuddin N, Al-Fuqaha A (2019) Blockchain for AI: review and open research challenges. IEEE Access 7:10127–10149. https://doi.org/10.1109/ACCESS.2018.2890507
Santana C, Albareda L (2022) Blockchain and the emergence of Decentralized Autonomous Organizations (DAOs): an integrative model and research agenda. Technol Forecast Soc Chang 182:121806. https://doi.org/10.1016/j.techfore.2022.121806
Schädler L, Lustenberger M, Spychiger F (2023) Analyzing decision-making in blockchain governance. Front Blockchain 6:1256651
Schuppli B, Jafari GA (2021) Piercing the Digital Veil: A Case Study for a DAO Legal Framework under Swiss Law, Vol. 12 JIPITEC 331 para 1.
Schwaninger M (2009) Intelligent organizations: powerful models for systemic management. Springer Verlag, Berlin. https://doi.org/10.1007/978-3-540-85162-2
Shrestha YR, Ben-Menahem SM, von Krogh G (2019) Organizational decision-making structures in the age of artificial intelligence. Calif Manage Rev 61(4):66–83. https://doi.org/10.1177/0008125619862257
Spychiger F, Tessone CJ, Zavolokina L, Schwabe G (2022) Incentivizing data quality in blockchain-based systems – the case of the digital cardossier. Distributed Ledger Technol Res Practice 1(1):1–27
Spychiger F, Makode PK, Küng L, Tessone CJ (2024) Governance and maintenance for a DAO with physical assets-an agent-based Model. In: 2024 IEEE international conference on omni-layer intelligent systems (COINS), pp 1–6
Spychiger F, Lustenberger M, Küng L (2025) Decision-Making in Decentralized Autonomous Organizations (DAOs). In: Themistocleous M (ed) Handbook of blockchain technology. Edward Elgar Publishing, UK
Tan JZ, Merk T, Hubbard S, Oak ER, Pirovich J, Rennie E et al (2023) Open problems in DAOs, arXiv preprint arXiv:2310.19201. https://doi.org/10.48550/arXiv.2310.19201
Vahs D (2019) Organisation: Ein Lehr- und Managementbuch, 10th edn. Schäffer-Poeschel Verlag, Stuttgart
Van Vulpen P, Jansen S (2023) Decentralized autonomous organization design for the commons and the common good. Front Blockchain 6:1287249
Vergne J (2020) Decentralized vs. distributed organization: blockchain, machine learning and the future of the digital platform. Org Theory 1(4):1–26. https://doi.org/10.1177/2631787720977052
Wang S, Ding W, Li J, Yuan Y, Ouyang L, Wang FY (2019) Decentralized autonomous organizations: concept, model, and applications. IEEE Trans Comput Soc Syst 6:870–878. https://doi.org/10.1109/TCSS.2019.2938190
Wright A (2021) The rise of decentralized autonomous organizations: opportunities and challenges. Stanford J Blockchain Law Policy 4(2):152–176
Ziolkowski R (2021) Exploring Blockchain Governance (Dissertation). https://www.zora.uzh.ch/id/eprint/211448/1/Dissertation_Ziolkowski_final.pdf. Accessed 21 Nov 2023
Ziolkowski R, Miscione G, Schwabe G (2020) Decision problems in blockchain governance: old wine in new bottles or walking in someone else's shoes? J Manag Inf Syst 37:316–348. https://doi.org/10.1080/07421222.2020.1759974

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